

NCERT Class 10 Science Doubt-Solving Assistant with Multi-Guard Semantic Cache

- Classification: Routes to STANDALONE, FOLLOW_UP, or STYLE using zero-LLM rules. First-turn follow-ups or style requests return helpful prompts with 0 LLM calls.
- Follow-Up Rewrite: Ambiguous follow-ups are rewritten to standalone questions via LLM; self-contained queries are preserved unchanged.
- Retrieval: Fetches top 6 chunks from 734 textbook chunks via FAISS using bge-small-en-v1.5 embeddings.
- Out-of-Scope Borderline Band: Score < 0.58 declines with 0 LLM calls; 0.58 to 0.68 lets the LLM inspect context; >= 0.68 answers directly.
- Generation: LLM answers strictly using context; numericals route to gpt-oss-120b with sign-convention prompts; failover on 429 to fallback model.

- Latency: Exact hit ~5 ms end-to-end (0.08 ms store); semantic hit ~11 ms (5.8 ms store); fresh LLM ~0.6s to 2s.
- Exact Match: Normalized queries (lowercased, punctuation-stripped) perform an instant SQLite lookup.
- Semantic Candidate Retrieval: Non-numerical queries search a FAISS vector index (cosine similarity ≥ 0.88).
- Five Safety Guards: Number Guard (identical values/units), Symbol Guard (exact match on letters C, F, P, 2F), Contrast Guard (concave/convex, acids/bases), Key-Term Guard (Jaccard ≥ 0.65), and Question-Type Guard (define vs law vs numerical).

- CACHED: Verified standalone NCERT answers with citations, canonical forms of follow-up questions, and contextual follow-up pairs.
- NEVER CACHED: Style restylings, out-of-scope declines, greetings, and numerical/calculation questions (containing digits or calculate/compute triggers) to prevent serving stale arithmetic.

- The 12 cm vs 60 cm Lens Answer Caching: A convex lens problem ($f=20\text{cm}$, $u=-30\text{cm}$) was answered with mirror formula ($1/v+1/u=1/f$) giving $v=+12\text{cm}$ instead of lens formula ($v=+60\text{cm}$). Because numericals were cached, this flawed answer was stored and served repeatedly. Fixed by routing numericals to gpt-oss-120b with sign conventions and strictly forbidding caching for numericals.
- Guessed 0.50 Scope Threshold vs Real Calibration: A 0.50 cutoff was initially guessed, but off-topic queries scored up to 0.57 (renewable energy, forests) while textbook topics scored around 0.61 (saponification) and 0.66 (neurons). Fixed by a calibrated 3-tier band: <0.58 declines with 0 LLM calls, 0.58 to 0.68 lets LLM decide, and ≥ 0.68 answers directly.
- Pure Cosine Similarity: Opposing concepts scored deceptively high (concave vs convex: 0.9505; R=20 vs R=30: 0.9244). Required explicit contrast, symbol, and number guards to prevent factual hallucinations.
- Strict Jaccard Thresholds: A 0.80 Jaccard threshold caused false misses on natural phrasing variations; lowering to 0.65 preserved safety while allowing natural phrasing.

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graph TD
    Start([User sends message]) --> Classify{classify_message  
rule-based, no LLM}
    Classify --> Style{STYLE}
    Classify --> FollowUp{FOLLOW-UP}
    Classify --> Standalone{STANDALONE}
    Style --> CacheLookup[cache_lookup  
skip cache]
    FollowUp --> CacheLookup
    Standalone --> CacheLookup
    CacheLookup --> CacheHit{Cache Hit}
    CacheHit -- "FOLLOW-UP: contextual  
exact match only" --> GenerateFromContext[generate LLM answers from context]
    CacheHit -- "STANDALONE: exact -  
semantic - guards" --> ScopeCheck{scope_check  
best score > threshold?}
    CacheHit -- MISS --> RewriteQuestion[rewrite_question  
LLM: standalone form]
    RewriteQuestion --> RetrieveFAISS[retrieve FAISS book index, top 4]
    ScopeCheck -- Yes --> GenerateFromContext
    ScopeCheck -- No --> GenerateNewAnswer[generate rewrite previous answer]
    GenerateFromContext --> Validate[validate  
check citations]
    GenerateNewAnswer --> Validate
    RewriteQuestion -- "STYLE: always skip" --> StandaloneOutput[STANDALONE]
    Validate --> MaybeStore{maybe_store  
cache policy check}
    MaybeStore -- Cacheable --> StoreInCache[Store in SQLite = FAISS]
    MaybeStore -- Not cacheable --> Finalize[finalize  
return cached answer]
    StoreInCache --> Finalize
    Finalize --> ReturnResponse[Return response  
reply citations, cache hit,  
latency_mis]

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